# POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
Course: Applied Chemistry

Semester : Spring

Year : 2023
Full Marks: 100
Pass Marks: 45
Time : 3hrs.

*Candidates are required to give their answers in their own words as far as practicable.*

*The figures in the margin indicate full marks.*

***Attempt all the questions.***

1. a) What are the differences between a galvanic cell and electrolytic cell? A galvanic cell consists of two electrode, zinc and lead, zinc electrode is immersed in 0.1 M zinc ion solution and lead electrode in 0.02 M lead ion solution. Calculate the emf of the cell at 298 K. Write the equations of electrode process and represent the cell schematically. Given that $E^0_{(Zn^{++}/Zn)}$= - 0.76 volt and $E^0_{(Pb^{++}/Pb)}$= - 0.13 volt — 8

**OR**

Define electrochemical series. Write the applications of electrochemical series. From the following data, calculate the emf of cell at $25^0$C.

$E^0\ Ni^{2+}/Ni$ = - 0.25V

$E^0\ Cu^{2+}/Cu$ = + 0.34V

$[Ni^{++}]$ = 0.1M, $[Cu^{2+}]$ = 0.2M

R = 8.314 J $mol^{-1}$ $K^{-1}$

F = 96500C

b) Explain primary and secondary Cells. Write the construction and working mechanism of lithium-ion battery. — 7

2. a) What is air pollution? Write its cause effect and control measure? — 7

b) What is hard water? How can you determine alkalinity of water experimentally? — 8

**OR**

What is permanent hardness of water? How can you determine total hardness of water experimentally

3. a) Explain the properties of transition metals with regards to the magnetic properties and variable oxidation states. — 7



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b) Give reasons:
   i. Transition metals are good in forming complexes.
   ii. $TiO_2$ is colourless.
   iii. Zn is not considered as true transition element.
   iv. Transition metals exhibit catalytic properties.

4. a) Define Nucleophilic Substitution reaction. Show the reaction mechanism between (R)-2-Bromobutane with water 8

   b) Differentiate $E_1$ and $E_2$ reactions with mechanism and kinetics. 7

5. a) Write functions of various constituents of paints. 5

   b) Write Principle and applications of sensors. 5

   c) Write Preparation, properties and uses of TNT. 5

6. a) Write short notes on addition polymer and condensation polymer. 8

   b) How do biodegradable polymers differ from non-biodegradable polymer? Explain with suitable examples. 7

7. Write short notes on: (Any two) 2×5

   a) Chemistry of cement

   b) Acidic and Basic Buffer

   c) Stereochemistry

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